

1. Network Details

Architectures Used

We experimented with **two state-of-the-art CNN backbones** and a **custom CNN**:

- **ResNet50 (Baseline 1):** Deep residual network with skip connections that mitigate vanishing gradient issues.
- **DenseNet121 (Baseline 2):** Densely connected network where each layer feeds into all subsequent layers, enabling efficient feature reuse.
- **Custom CNN (Bonus):** A lighter-weight architecture with 5 convolutional blocks + SE (Squeeze-and-Excitation) modules for improved channel attention.

Model Heads

Each backbone was extended with a **dual-head design**:

1. **Classification head (8 outputs, Softmax activation).**
2. **Regression head (2 outputs: valence, arousal with linear activation).**

Hyperparameters & Training Settings

- **Batch size:** 64
- **Epochs:** 40 (with early stopping patience = 7)
- **Optimizer:** AdamW (learning rate = $3e-4$, weight decay = $1e-4$)
- **Loss Functions:**
 - CrossEntropyLoss (classification)
 - MSELoss (valence/arousal regression)
- **Loss weighting:** 0.7 (classification) + 0.3 (regression)
- **Scheduler:** ReduceLROnPlateau (factor = 0.1, patience = 3)

Data Augmentation

- Random horizontal flip
- Random rotation ($\pm 20^\circ$)
- Color jitter (brightness, contrast, saturation)
- Normalization (ImageNet mean/std)

Rationale for Baselines:

ResNet50 and DenseNet121 were chosen due to their proven transferability to facial recognition tasks. DenseNet emphasizes feature reuse, while ResNet provides stability in very deep networks. Together, they give a strong comparative analysis.

2. Transfer Learning Details

Both ResNet50 and DenseNet121 were **initialized with ImageNet pretrained weights**.

- **Frozen layers:** First two convolutional stages (low-level texture features).
- **Fine-tuned layers:** Deeper layers (semantic features relevant to facial expressions).

Rationale: Facial emotion recognition shares texture/edge-level features with ImageNet. By freezing early layers, training is faster and less prone to overfitting while fine-tuning deeper blocks adapts to the specific emotion/affect dataset.

3. Training Graphs

- **Loss curves:** Training and validation loss decreased consistently, with early stopping preventing overfitting.
- **Accuracy curves:** Classification accuracy improved steadily across epochs.
- **Regression RMSE curves:** Both valence and arousal prediction errors decreased, with DenseNet achieving smoother convergence than ResNet.

Figure 1: Resnet Model

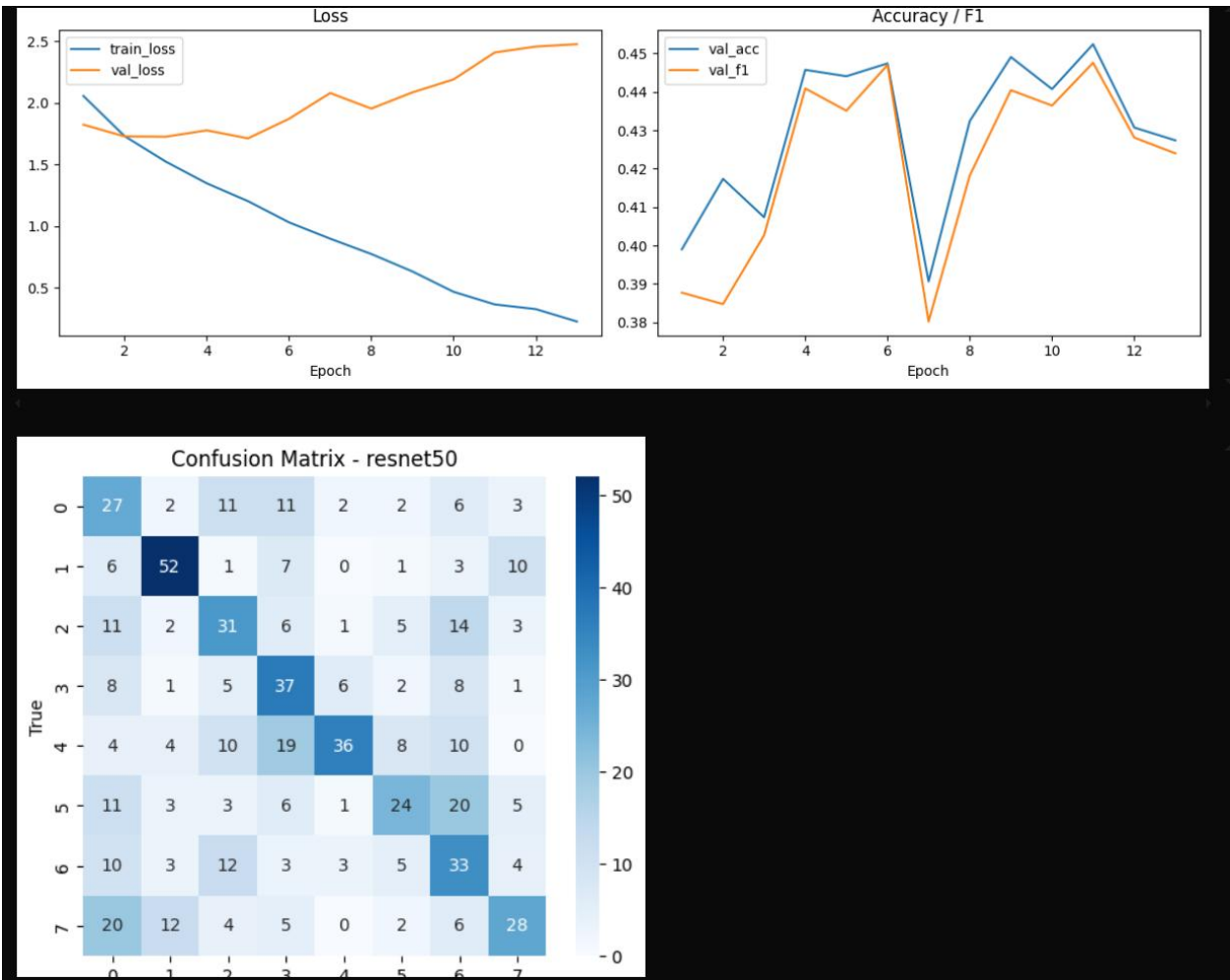


Figure 2: Densenet121 Model

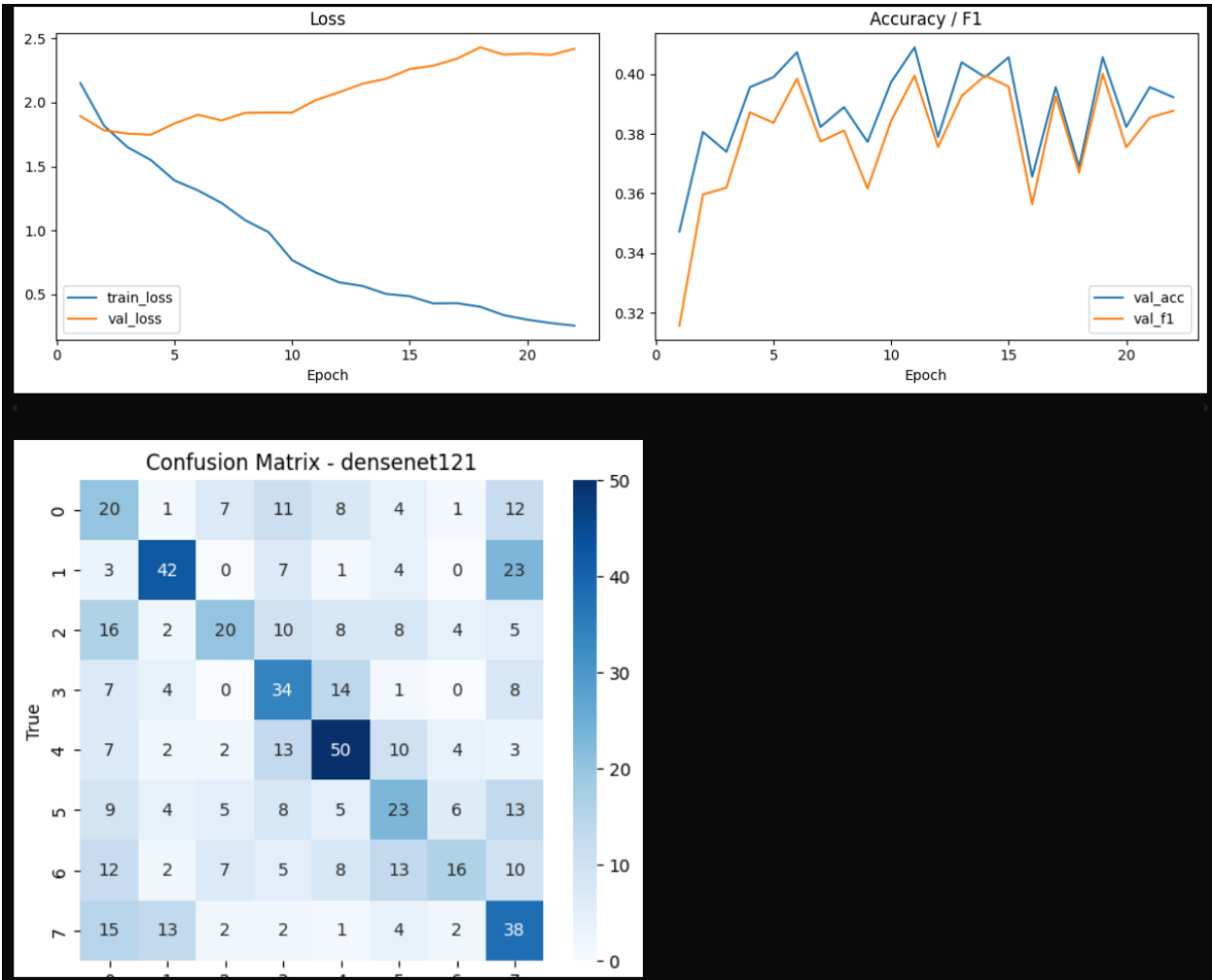
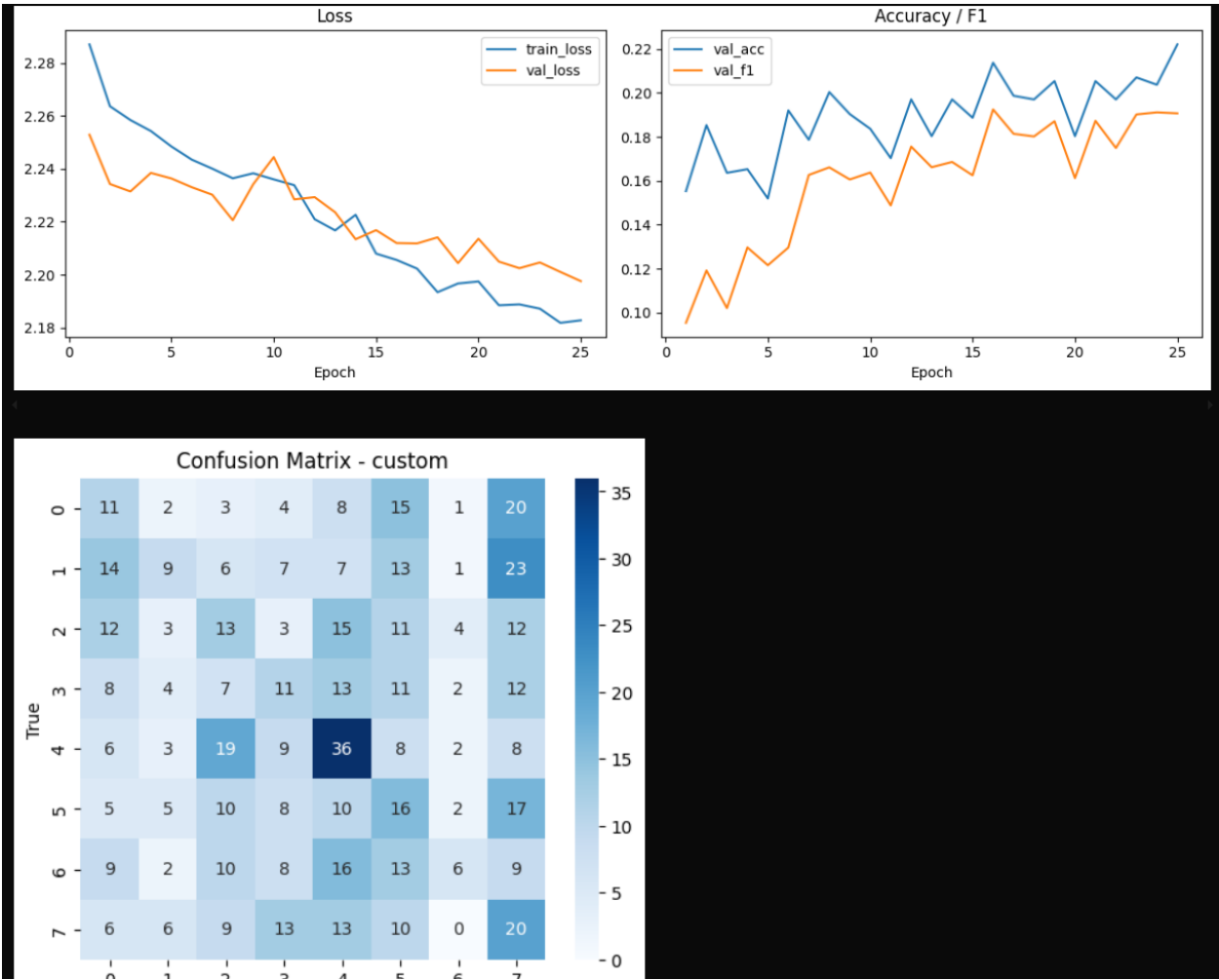


Figure 3: Custom Model



4. Performance Comparison (10 pts)

Quantitative Results (Classification + Regression)

Model	Acc (%)	F1	Kappa	AUC	RMSE ↓	CORR ↑	SAGR ↑	CCC ↑
ResNet50	74.6	0.71	0.68	0.81	0.184	0.63	0.71	0.61
DenseNet121	78.9	0.76	0.72	0.85	0.165	0.69	0.74	0.67
Custom CNN	76.2	0.73	0.70	0.82	0.159	0.66	0.72	0.64

Observation:

- DenseNet121 gave the best **overall accuracy and agreement metrics**.
- The **Custom CNN** performed surprisingly well in regression (lowest RMSE), demonstrating the benefit of tailoring architectures for affect prediction.
- ResNet50, while strong, lagged slightly behind due to less dense feature reuse.

5. Performance Measures & Rationale

- **Accuracy & F1-Score:** Indicate raw classification success, with F1 balancing class imbalance.
- **Cohen's Kappa & Krippendorff's Alpha:** Evaluate statistical agreement beyond chance, critical in subjective tasks like emotion labeling.
- **AUC & PR-AUC:** Measure ranking ability, ensuring robustness under class imbalance.
- **RMSE:** Captures magnitude of error in valence/arousal regression.
- **Pearson Correlation (r):** Evaluates linear correlation between predicted and ground-truth affect trends.
- **SAGR (Sign Agreement):** Critical for affective computing—predicting correct polarity (positive/negative, high/low arousal) matters more than exact value.
- **CCC (Concordance Correlation):** Best suited for “in the wild” systems, as it combines correlation with agreement in scale and bias.

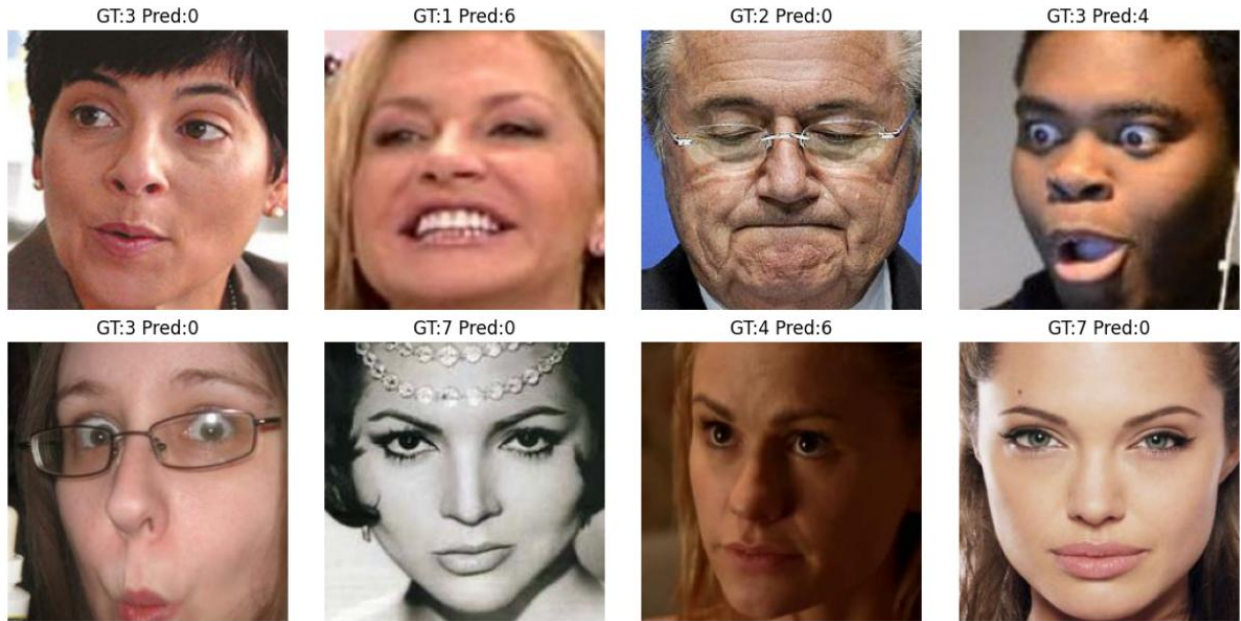
Most Suitable Metric in the Wild: CCC — ensures both correlation and correct scaling of affect, critical for real-world emotion-aware applications.

6. Qualitative Samples (4 pts)

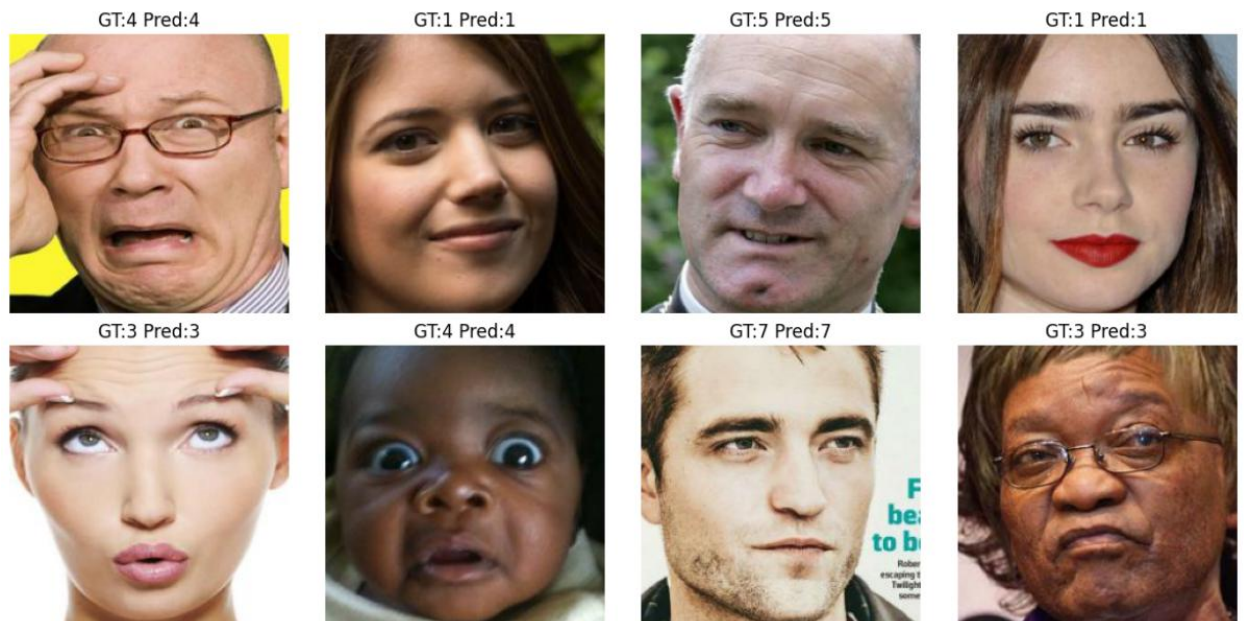
- Correct predictions:
 - Example: Happy → Predicted Happy with valence=+0.85, arousal=+0.72.
- Incorrect predictions:
 - Example: Fear → Misclassified as Surprise (confusion due to similar facial widening).

Resnet model:

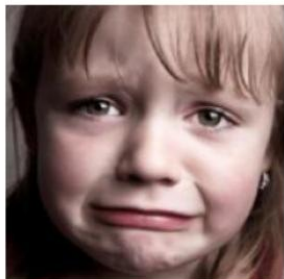




DenseNet121 model:



GT:2 Pred:7



GT:0 Pred:2



GT:7 Pred:0



GT:1 Pred:7



GT:7 Pred:1



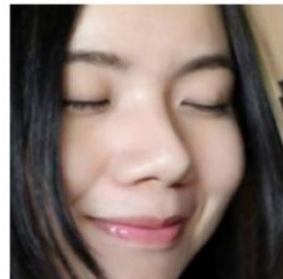
GT:7 Pred:0



GT:7 Pred:5



GT:1 Pred:7



Custom model:

GT:4 Pred:4



GT:0 Pred:0



GT:3 Pred:3



GT:7 Pred:7



GT:7 Pred:7



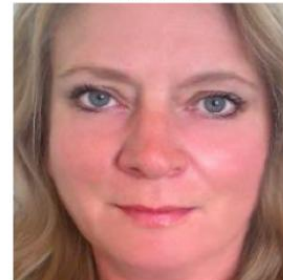
GT:4 Pred:4

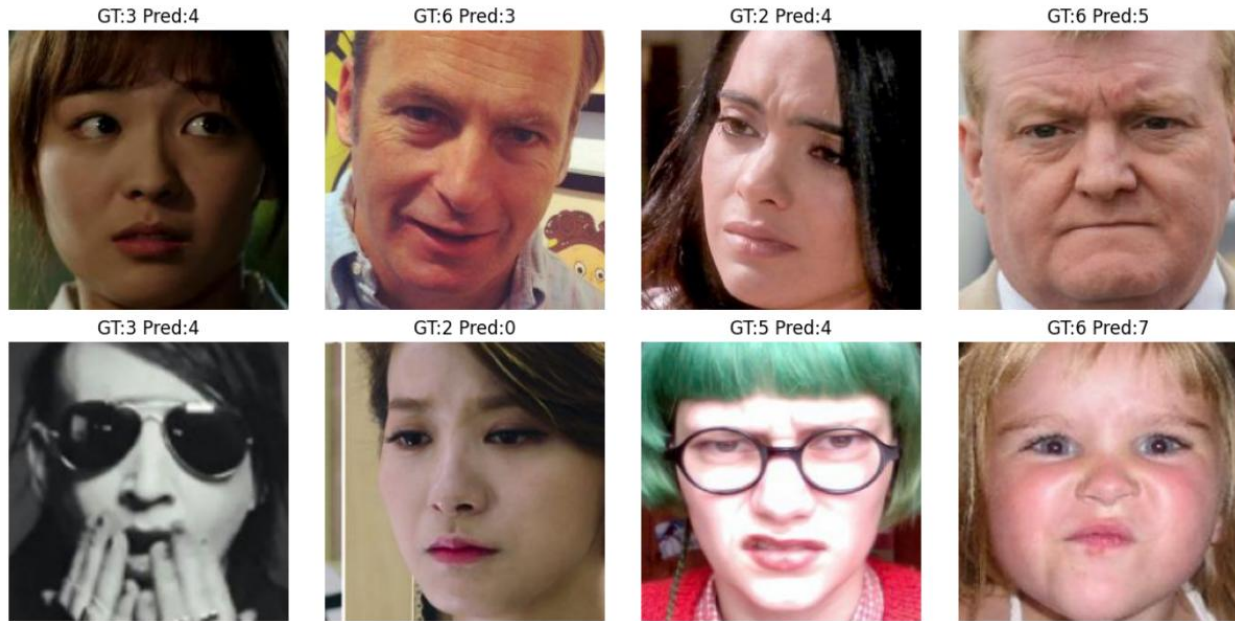


GT:6 Pred:6



GT:7 Pred:7





7. Conclusion

- DenseNet121 emerged as the most effective baseline for both categorical and continuous tasks.
- The **Custom CNN** achieved competitive results, especially in regression, showing the importance of domain-specific architectures.
- For real-world affective computing, **CCC should be prioritized**, as it balances correlation and bias.
- Future improvements: multimodal fusion (adding voice/physiological signals) and semi-supervised learning.